Final Module Task - Piechaczek

```
# liczę konta w rozbiciu na datę, kraj, send_interval, is_verified, is_unsubscribed:
```

Code:

```
WITH account_metrics AS (
SELECT
COUNT(DISTINCT a.id) AS account_cnt,
s.date,
sp.country,
a.send_interval,
a.is_verified,
a.is_unsubscribed,
NULL AS email_sent,
NULL AS email_open,
NULL AS email_visit
FROM data-analytics-mate.DA.account AS a
JOIN data-analytics-mate.DA.account_session AS acs
ON a.id = acs.account_id
JOIN data-analytics-mate.DA.session_params AS sp
ON acs.ga_session_id = sp.ga_session_id
JOIN data-analytics-mate.DA.session AS s
ON acs.ga_session_id = s.ga_session_id
GROUP BY
s.date,
sp.country,
a.send_interval,
a.is_verified,
a.is_unsubscribed
),
# w kolejnym CTE liczę liczbę wysłanych, otwartych i klikniętych maili z takim samym
grupowaniem jak powyżej:
email_metrics AS(
 SELECT
 NULL AS account_cnt,
 DATE_ADD(s.date,INTERVAL es.sent_date DAY) AS sent_date,
 sp.country,
```

```
 a.send_interval,
 a.is_verified,
 a.is_unsubscribed,
 COUNT(DISTINCT es.id_message) AS email_sent,
 COUNT(DISTINCT eo.id_message) AS email_open,
 COUNT(DISTINCT ev.id_message) AS email_visit
 FROM data-analytics-mate.DA.email_sent AS es
 LEFT JOIN data-analytics-mate.DA.email_open AS eo
 ON es.id_message = eo.id_message
 LEFT JOIN data-analytics-mate.DA.email_visit AS ev
 ON es.id_message = ev.id_message
 LEFT JOIN data-analytics-mate.DA.account_session AS acs
 ON es.id_account = acs.account_id
 JOIN data-analytics-mate.DA.account AS a
 ON a.id = acs.account_id
 JOIN data-analytics-mate.DA.session_params AS sp
 ON acs.ga_session_id = sp.ga_session_id
 JOIN data-analytics-mate.DA.session AS s
 ON acs.ga_session_id = s.ga_session_id
 GROUP BY
 sent_date,
 sp.country,
 a.send_interval,
 a.is_verified,
 a.is_unsubscribed
),
# łączę obie tabele w jedną za pomocą UNION ALL:
union_metrics AS (
SELECT *
FROM account_metrics

UNION ALL

SELECT *
FROM email_metrics
),
```

```
# żeby dostać finalną tabelę, sumuję liczbę kont, wysłane, otwarte i kliknięte maile po grupowaniu: date, country, send_interval, is_verified, is_unsubscribed. Dla bezpieczeństwa używam funkcji COALESCE() która zastąpi wartości NULL inną wartością (np.0):
message_cnt AS (
SELECT
date,
country,
send_interval,
is_verified,
is_unsubscribed,
SUM(COALESCE(account_cnt, 0)) AS account_cnt,
SUM(COALESCE(email_sent, 0)) AS sent_msg,
SUM(COALESCE(email_open, 0)) AS open_msg,
SUM(COALESCE(email_visit, 0)) AS visit_msg
FROM union_metrics
GROUP BY
date,
country,
send_interval,
is_verified,
is_unsubscribed
ORDER BY
date,
country,
send_interval
),

# za pomocą funkcji okienkowych obliczam: łączną liczbę kont w kraju, łączną liczbę wysłanych maili w kraju:
final AS(
SELECT
*,
SUM(account_cnt) OVER(PARTITION BY country) AS total_country_account_cnt,
SUM(sent_msg) OVER(PARTITION BY country) AS total_country_sent_cnt
FROM message_cnt
```

```
),

# używając funkcji DENSE_RANK() nadaję krajom miejsce w rankingu według łącznej liczby
kont malejąco:
ranking AS (
SELECT
*,
DENSE_RANK() OVER(ORDER BY total_country_account_cnt DESC) AS
rank_total_country_account_cnt,
DENSE_RANK() OVER(ORDER BY total_country_sent_cnt DESC) AS rank_total_country_sent_cnt
FROM final
)
# filtruję top 10 (records where rank_total_country_account_cnt or rank_total_country_sent_cnt is
less than or equal to 10):
SELECT *
FROM ranking
WHERE rank_total_country_account_cnt <= 10 OR rank_total_country_sent_cnt <= 10
ORDER BY
rank_total_country_account_cnt,
rank_total_country_sent_cnt;
```



Job information | Results | Visualization | JSON | Execution details | Execution graph

w	date	country	send_interval	is_verified	is_unsubscribed	account_cnt	sent_msg	open_msg	visit_msg	tot
1	2020-12-26	United States	7	1	1	0	6	1	0	
2	2020-12-17	United States	1	0	0	54	37	15	0	
3	2021-01-29	United States	7	1	1	0	5	0	0	
4	2020-11-08	United States	1	0	0	18	17	5	0	
5	2020-12-03	United States	1	1	1	25	210	120	18	
6	2020-11-10	United States	1	0	1	2	0	0	0	
7	2020-11-18	United States	7	1	1	0	9	9	2	
8	2021-01-21	United States	3	1	0	0	3	3	0	
9	2021-01-12	United States	1	1	0	99	2384	751	63	
10	2020-12-29	United States	7	1	1	3	8	2	1	
11	2021-01-26	United States	1	0	0	37	19	8	0	
12	2020-12-17	United States	1	1	1	37	269	160	18	
13	2020-12-25	United States	7	1	1	0	6	0	0	
14	2021-02-14	United States	1	1	0	0	875	330	26	
15	2021-01-18	United States	7	1	0	0	15	10	1	

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account_cnt

21,327







